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COMMERCIAL ITEM DESCRIPTION

WELDING TORCHES AND COMPONENT PARTS, GAS TUNGSTEN ARC WELDING (GTAW)

The General Services Administration has authorized the use of this commercial item description.

1. ABSTRACT

1.1 Abstract. This description covers welding torches which are used for gas tungsten arc welding (GTAW), commonly referred to as tungsten inert gas (TIG) welding.

2. CLASSIFICATION

2.1 Classification. The torches shall be of the following types and sizes. The type, size, and quantity of torches to be furnished shall be as specified (see 7.1).

Type I - Air Cooled		Type II - Water Cooled	
Size	Current (amperes)	Size	Current (amperes)
75	75	125	125
100	100	180	180
150	150	200	200
200	200	250	250
		300	300
		350	350
		500	500
		650	650

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis TN 38114-5213.

AMSC N/A FSC 3431

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3. SALIENT CHARACTERISTICS

3.1 Design. Unless otherwise specified (see 7.1), the torch furnished shall be designed to be interchangeable with what is referred to in the industry as a Linde style torch. This includes collets, collet bodies, back caps, gas lenses, nozzles, and all other component parts. When specified, the torch furnished shall be interchangeable with an Airco style torch.

3.2 Torch body and handle. The current carrying components of the torch shall be made of high conductive copper. The exterior of the torch body and handle shall be constructed of a lightweight, high impact resistant, nonconductive material. Unless otherwise specified (see 7.1), the torch head shall be fixed at a 60 to 75 degree angle when measured from the center line of the handle to the center line of the backcap. When specified (see 7.1), the handle shall have a fixed head with a 90 or 180 degree angle, as specified, or a flexible neck for operator positioning of the head angle. When required, the torch body shall be furnished with cup gasket. The type I torch shall be furnished with all necessary fittings for connecting to a gas hose and electrical power cable assembly. The type II shall be furnished with all necessary fittings for connecting to a gas hose, water hose, and electrical power cable assembly.

3.2.1 Duty cycle. The type I torches furnished shall be rated at not less than 60 percent duty cycle and the type II shall be rated at 100 percent duty cycle. Duty cycle is defined as the ratio of arc time to total time. The time period of one complete test cycle shall be 10 minutes.

3.2.2 Gas valve. When specified by the procuring activity (see 7.1), the torch body shall be furnished with a built in gas valve to provide on-off control of shielding gas flow.

3.2.3 Marking. The torch body shall be permanently identified with the manufacturer's name or trademark and the manufacturer's model or part number.

3.3 Type I, power cable, and gas hose assembly. When specified (see 7.1), the torch shall be furnished with a power cable and gas hose assembly. The operating pressure of the gas hose shall be not less than 150 pounds per square inch gauge (psig). Unless otherwise specified (see 7.1), the assembly shall be not less than 12.5 feet in length and shall be furnished complete with all necessary connectors. When specified (see 7.1), a protective cover shall be furnished for the assembly. The protective cover shall be of sufficient length to cover from 70 to 90 percent of the cable length.

3.4 Type II, power cable, gas hose, and water hose assembly. When specified (see 7.1), the torch shall be furnished with a power cable, gas hose, and water hose assembly. The operating pressure of the gas and water hose shall be not less than 150 psig. Unless otherwise specified (see 7.1), the assembly shall be not less than 12.5 feet in length and shall be furnished complete with all necessary connectors. When specified (see 7.1), a protective cover shall be furnished for the assembly. The protective cover shall be of sufficient length to cover from 70 to 90 percent of the assembly length.

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3.5 Collets, collet bodies, and gas lens collet Bodies.

3.5.1 Collets. When specified (see 7.1), a collet or collets shall be furnished. The size and quantity of collets to be furnished shall be as specified (see 7.1) and shall be selected from the following sizes (diameters). Collet size is determined by the diameter of electrode with which it will be used.

0.020 inch	1/8 inch
0.040 inch	5/32 inch
1/16 inch	3/16 inch
3/32 inch	1/4 inch

3.5.2 Collet bodies. When specified (see 7.1), all necessary collet bodies shall be furnished to cover all sizes of collets selected in 3.5.1. Unless otherwise specified (see 7.1), the collet bodies furnished shall be the manufacturer's standard item. The collet bodies shall be furnished with any additional gaskets or insulators which may be required. Collet bodies which cover a range of collets are acceptable.

3.5.3 Gas lens collet bodies. When specified, (see 7.1), gas lens collet bodies shall be furnished. The size of the gas lens to be furnished shall be as specified (see 7.1), and shall be selected from the list of sizes in 3.5.1. Unless otherwise specified (see 7.1), the gas lens furnished shall be the manufacturer's standard item. The gas lens shall be furnished with any additional gasket or insulator which may be required. Gas lens collet bodies which cover a range of collets are acceptable.

3.6 Gas Cups. When specified (see 7.1), all necessary gas cups to fit the collet bodies and gas lens collet bodies requested, in the solicitation or purchase order, shall be furnished.

3.6.1 Gas cup orifice size. Unless otherwise specified, the orifice diameter of the gas cup(s) to be furnished shall be as specified by the procuring activity (see 7.1), and selected from the following sizes (diameters):

3/16 inch	3/8 inch	5/8 inch
1/4 inch	7/16 inch	11/16 inch
5/16 inch	1/2 inch	3/4 inch

3.6.2 Gas cup material. Unless otherwise specified, the gas cup(s) furnished shall be manufactured from alumina. When specified (see 7.1), the cups shall be manufactured from a material selected from the following list and shall be as specified by the procuring activity (see 7.1).

Lava
Metal
Metal shielded alumina

3.6.3 Gas cup length. Unless otherwise specified (see 7.1), the length of the gas cup shall be between 1 and 2 inches in length.

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3.7 Back cap. The back cap(s) furnished shall be complete with all necessary "O" rings or gaskets and shall be identified as short, medium, or long. A short backcap shall allow the use of an electrode not less than 2 inches in length, the medium backcap shall allow the use of an electrode not less than 3 inches in length, and a long backcap shall allow the use of an electrode not less than 7 inches in length. The length of the back caps furnished shall be as specified (see 7.1).

3.8 Tools. All special tools which are required for replacement of operator replaceable torch components shall be furnished.

3.9 Operating and maintenance instructions. Operating and maintenance instructions shall be provided with each torch outfit furnished. The instructions shall include directions and information necessary for the safe operation and maintenance of the outfit and include a list of replacement parts with part numbers. The instructions shall be written in the English language.

3.10 Replacement components. Each welding torch and or component shall be a replacement for the brand name and model number of torch as specified (see 7.1). All components shall be interchangeable with the brand name and model specified. Unless otherwise specified (see 7.1), the torch components ordered shall meet the requirements given in paragraphs 1 thru 7.

3.11 Optional components. Optional components shall be furnished as specified and fully described (see 7.1). The components shall be functional without requiring modification of the equipment or the torch.

3.12 General requirements.

3.12.1 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3.12.2 Environmental protection. During the manufacture of the torch, materials hazardous to the ecological system as prohibited by the Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

4. REGULATORY REQUIREMENTS

4.1 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

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5. QUALITY ASSURANCE PROVISIONS

5.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PRESERVATION, PACKAGING, PACKING, AND MARKING

6.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES.

7.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Type, size, and quantity required (see 2.1).

c. Specify Airco style torch or parts, if required (see 3.1).

d. Specify head angle, if different (see 3.2).

e. Specify type of head, if different (see 3.2).

f. Gas valve, if required (see 3.2.2).

g. Type I power cable and gas hose assembly, when required (see 3.3).

h. Type II power cable, gas hose, and water hose assembly, when required (see 3.4).

i. Length of power cable and hose assembly, if different (see 3.3 and 3.4).

j. Protective covering for power cable and hose, when required (see 3.3 and 3.4).

k. Specify collet size(s) and quantity, if required (see 3.5.1).

l. Collet bodies(s), if required (see 3.5.2).

m. Describe collet bodies, if different (see 3.5.2).

n. Specify gas lens collet bodies size and quantity, when required (see 3.5.3).

o. Describe gas lens collet bodies, if different (see 3.5.3).

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o. Gas cup(s) and quantity, when required (see 3.6).

p. Gas cup(s) sizes to be furnished (see 3.6.1).

q. Gas cup(s) material, if different (see 3.6.2).

r. Gas cup length, if different (see 3.6.3).

s. Specify length of back cap(s) and quantity required (see 3.7).

t. Specify brand name and part number of torch requiring replacement components (see 3.10).

u. Describe and state quantity of replacement components (see 3.10).

v. Specify and describe optional equipment required (see 3.11).

7.2 Source of reference documents.

Federal Standard FED-STD-376 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5099.

7.3 CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

A51187-X X

--------- Size: A - 75, D - 150, G - 250, J - 500
B - 100, E - 180, H - 300, K - 650
C - 125, F - 200, I - 350,

---------- Type: 1 - Type I, 2 - Type II

------------------- CID number

MILITARY INTEREST:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activities

Army - ME, CE
Air Force - 84
DLA - GS

User Activities

Navy - YD, SH, MC

CIVIL AGENCY COORDINATION ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3431-0221

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